

Computer Networks

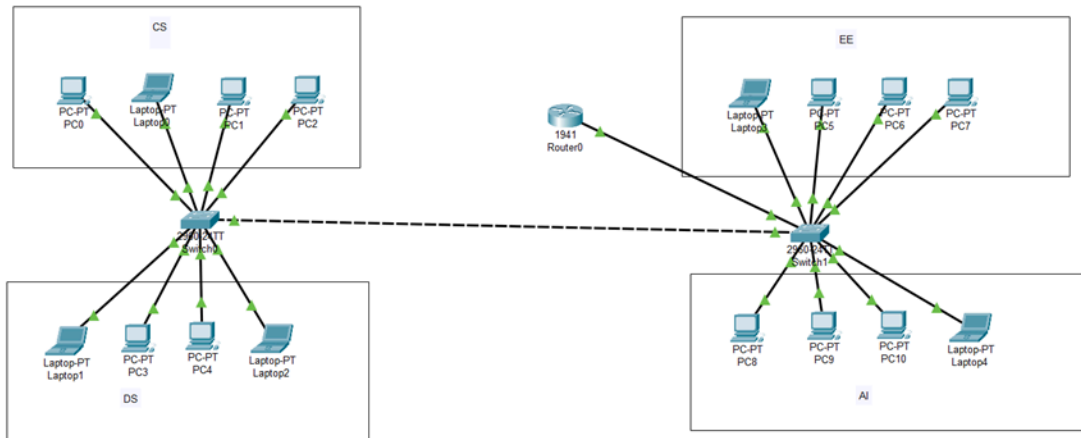
LAB#11



23K-2001

BCS-5J

Q1:



Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

VLAN Configuration

VLAN Number: 1

VLAN Name: default

Add Remove

VLAN No	VLAN Name
1	default
10	CS
20	DS
24	EE
40	AI

Switch1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

VLAN Configuration

VLAN Number: 40

VLAN Name: AI

Add Remove

VLAN No	VLAN Name
1	default
10	CS
20	DS
24	EE
40	AI

IOS Command Line Interface

```
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name DS
Switch(config-vlan)#vlan 24
Switch(config-vlan)# name EE
Switch(config-vlan)#vlan 40
Switch(config-vlan)# name AI
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
%SYS-5-CONFIG_I: Configured from console by console

Switch(config)#enable
% Incomplete command.
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range f0/1-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range f0/5-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range f0/9
Switch(config-if-range)#interface f0/9
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20,24,40
Switch(config-if)#
```

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name CS
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name DS
Switch(config-vlan)#vlan 24
Switch(config-vlan)# name EE
Switch(config-vlan)#vlan 40
Switch(config-vlan)# name AI
Switch(config-vlan)#vlan 40
Switch(config-vlan)# name AI
Switch(config-vlan)%%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/9

%%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/9 on VLAN0001. Inconsistent port type.

%%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down

%%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up

Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#%SYS-5-CONFIG_I: Configured from console by console

Switch#rname
Translating "rname"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range f0/1-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 24
Switch(config-if-range)#exit
Switch(config)#interface range f0/5-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#interface f0/9
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,24,40
Switch(config-if)#exit
Switch(config)#interface f0/10
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,24,40
Switch(config-if)#exit
Switch(config)#
```

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dotIQ 10
                                     ^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.24
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.24, changed state to up

Router(config-subif)#encapsulation dot1Q 24
Router(config-subif)#ip address 192.168.24.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shut
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Router#
```

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

PC6

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.24.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.24.1

DNS Server 0.0.0.0

